# 1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

**Started on** Thursday, 24 September 2026, 10:25 AM

**State** Finished

**Completed on** Thursday, 24 September 2026, 11:30 AM

**Time taken** 1 hour 4 mins

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** | Correct | Mark 1.00 out of 1.00 | Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
| --- | --- |
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    int A[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }
    int slow = A[0];
    int fast = A[0];
    do {
        slow = A[slow];
        fast = A[A[fast]];
    } while (slow != fast);
    slow = A[0];

    while (slow != fast) {
        slow = A[slow];
        fast = A[fast];
    }
    printf("%d\n", slow);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

# 2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

**Started on** Thursday, 24 September 2026, 10:27 AM

**State** Finished

**Completed on** Thursday, 24 September 2026, 11:30 AM

**Time taken** 1 hour 3 mins

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
| --- | --- |
| 5<br>1 1 2 3 4 | 1 |

```c
#include <stdio.h>
#include <stdlib.h>
int main() {
    int n;
    if (scanf("%d", &n) != 1)
    return 0;
    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    int tortoise = arr[0];
    int hare = arr[0];
    do {
        tortoise = arr[tortoise];
        hare = arr[arr[hare]];
    } while (tortoise != hare);
    tortoise = arr[0];
    while (tortoise != hare) {
        tortoise = arr[tortoise];
        hare = arr[hare];
    }
    printf("%d\n", tortoise);
    free(arr);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

# 3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Com

**Started on** Thursday, 24 September 2026, 10:28 AM

**State** Finished

**Completed on** Thursday, 24 September 2026, 11:31 AM

**Time taken** 1 hour 3 mins

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

void solve() {
    int n1;
    if (scanf("%d", &n1) != 1) return;
    int *a1 = (int *)malloc(n1 * sizeof(int));
    for (int i = 0; i < n1; i++) {
        scanf("%d", &a1[i]);
```

```
    int n2;
    scanf("%d", &n2);
    int *a2 = (int *)malloc(n2 * sizeof(int));
    for (int i = 0; i < n2; i++) {
        scanf("%d", &a2[i]);
    }

    int i = 0, j = 0;
    int first = 1;
    while (i < n1 && j < n2) {
        if (a1[i] == a2[j]) {
            if (!first) printf(" ");
            printf("%d", a1[i]);
            first = 0;
            i++;
            j++;
        } else if (a1[i] < a2[j]) {
            i++;
        } else {
            j++;
        }
    }
    printf("\n");

    free(a1);
    free(a2);
}

int main() {
    int t;
    if (scanf("%d", &t) == 1) {
        while (t--) {
            solve();
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1 | 10 57 | 10 57 | ✔ |

```
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

**Correct**

Marks for this submission: 1.00/1.00.

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# 3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Co

**Started on** Thursday, 24 September 2026, 10:28 AM

**State** Finished

**Completed on** Thursday, 24 September 2026, 11:31 AM

**Time taken** 1 hour 3 mins

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

Sample:

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
| --- | --- |
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

void solve() {
    int n1;
    if (scanf("%d", &n1) != 1) return;
    int *a1 = (int *)malloc(n1 * sizeof(int));
    for (int i = 0; i < n1; i++) {
        scanf("%d", &a1[i]);
    }

    int n2;
    scanf("%d", &n2);
    int *a2 = (int *)malloc(n2 * sizeof(int));
    for (int i = 0; i < n2; i++) {
        scanf("%d", &a2[i]);
    }
```

```
    int i = 0, j = 0;
    int first = 1;
    while (i < n1 && j < n2) {
        if (a1[i] == a2[j]) {
            if (!first) printf(" ");
            printf("%d", a1[i]);
            first = 0;
            i++;
            j++;
        } else if (a1[i] < a2[j]) {
            i++;
        } else {
            j++;
        }
    }
    printf("\n");

    free(a1);
    free(a2);
}

int main() {
    int t;
    if (scanf("%d", &t) == 1) {
        while (t--) {
            solve();
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6 | 1 6 | 1 6 | ✔ |

```
45          }
46      }
47      return 0;
48  }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

**Correct**

Marks for this submission: 1.00/1.00.

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# 4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Comp

**Started on** Thursday, 24 September 2026, 10:29 AM

**State** Finished

**Completed on** Thursday, 24 September 2026, 11:31 AM

**Time taken** 1 hour 2 mins

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
| --- | --- |
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

void solve() {
    int n1;
    if (scanf("%d", &n1) != 1) return;
    int *a1 = (int *)malloc(n1 * sizeof(int));
    for (int i = 0; i < n1; i++) {
        scanf("%d", &a1[i]);
    }

    int n2;
    scanf("%d", &n2);
    int *a2 = (int *)malloc(n2 * sizeof(int));
    for (int i = 0; i < n2; i++) {
        scanf("%d", &a2[i]);
    }

    int i = 0, j = 0;
    int first = 1;
    while (i < n1 && j < n2) {
        if (a1[i] == a2[j]) {
            if (!first) printf(" ");
            printf("%d", a1[i]);
            first = 0;
```

```
            if (!first) printf(" ");
            printf("%d", a1[i]);
            first = 0;
            i++;
            j++;
        } else if (a1[i] < a2[j]) {
            i++;
        } else {
            j++;
        }
    }
    printf("\n");

    free(a1);
    free(a2);
}

int main() {
    int t;
    if (scanf("%d", &t) == 1) {
        while (t--) {
            solve();
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

# 5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

**Started on** Thursday, 24 September 2026, 11:23 AM

**State** Finished

**Completed on** Thursday, 24 September 2026, 11:43 AM

**Time taken** 20 mins

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    long long *A = (long long *)malloc(n * sizeof(long long));
    for (int i = 0; i < n; i++) {
        scanf("%lld", &A[i]);
    }
    long long k;
    scanf("%lld", &k);
    int i = 0, j = 1;
    int found = 0;
    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }
        long long diff = A[j] - A[i];
        if (diff == k) {
            found = 1;
            break;
        } else if (diff < k) {
            j++;
        } else {
            i++;
            if (i == j) {
                j++;
            }
        }
    }
    printf("%d\n", found);
    free(A);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

**Correct**

Marks for this submission: 1.00/1.00.

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# 6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

**Started on** Thursday, 24 September 2026, 10:30 AM

**State** Finished

**Completed on** Thursday, 8 October 2026, 10:25 AM

**Time taken** 13 days 23 hours

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

| Input | Result |
| --- | --- |
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *a = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0, j = 1;
    int found = 0;

    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }

        int diff = a[j] - a[i];
        if (diff == k) {
            found = 1;
            break;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }
    }

```

```
            j++;
        } else {
            i++;
        }
    }

    printf("%d\n", found);

    free(a);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔